

Substrate Monism: One Algebra, Seven Vocabularies, All Forces (Paper 19)

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May 6, 2026

Abstract

Papers 6, 13–18 of the Null Worldtube Theory (NWT) program have derived the Standard Model parameters, the fine-structure constant α , the 24-particle hadron mass spectrum, the gravitational coupling G , and the long-wavelength gravitational dynamics from a single substrate algebra: the octonion Clifford algebra $\text{Cl}(0, 7)$, the binary icosahedral group $2I$, the Brieskorn–Poincaré homology sphere $S^3/2I$, and the complete graph K_7 on seven vertices. Paper 18 [17], in particular, derives Einstein’s equations as the long-wavelength dynamics of the NWT condensate via Sakharov-induced gravity, and identifies the K_7 amplitude as the ultraviolet/infrared bridge that fixes Newton’s G to within 11 ppm of CODATA, inside the ± 22 ppm experimental uncertainty. This paper assembles the resulting *substrate-monism* thesis as a single, computationally verified demonstration.

The thesis is implemented in code. The open-source Python library `nwt_substrate` (344/344 unit tests passing) places the substrate primitives at its core and surrounds them with seven *view-shims*—QED, QCD, QFT, string theory, IBM Heron quantum hardware, electroweak, and gravity—each of which exposes the substrate through its community’s idiomatic vocabulary. The seven shims do not duplicate computations; they label the same Python objects. We walk a single electron through all seven lenses and verify that the same observable (mass, charge, magnetic moment, β_0 , Klein–Nishina amplitude, Z-pole cross section, G , m_e/m_{Pl}) computed from each lens agrees to the precision of the underlying derivation. Heron Experiment 10, executed on `ibm.marrakesh` on 2026-05-01, runs muon decay on the $|K_7\rangle$ substrate vacuum at the 9-qubit, 40-gate, depth-13 level and produces the predicted $\cos^2(\theta)$ survival curve at 96.3% contrast with 1.6% RMS deviation from theory—the K_7 that hosts the gravitational Wilson amplitude is, on real superconducting hardware, the substrate vacuum of the V–A weak vertex.

We close by displaying the master table: every entry of the Standard-Model + gravitational column—particle masses, gauge couplings, α , σ at the Z pole, Γ_Z , G , m_e/m_{Pl} , the Planck–electron mass hierarchy, the Klein–Nishina amplitude, the muon-decay survival curve on real hardware—is computed from a small set of substrate primitives within `nwt_substrate`. We treat substrate monism as a *computational hypothesis*: a commitment that one library, one set of substrate primitives, and no row-specific tuning suffice to reproduce the table. Of the 22 rows, 5 are independent structural predictions, 4 are direct hardware measurements, 6 are textbook QED/QCD identities reproduced from the substrate algebra, 5 are inherited from prior NWT papers, and 2 are algebraic corollaries; the framework’s empirical exposure rests on the 9 independent + hardware rows together with the operational consistency of the remaining 13. Whether this computational unification reflects a single physical substrate or only a sophisticated common interface is the program’s central interpretive question, to which the present paper supplies the executable evidence.

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1 Introduction

1.1 The substrate-monism thesis

Null Worldtube Theory (NWT) makes one structural claim about the physical world. There is a single underlying mathematical object, the *substrate*, and the various sub-disciplines of theoretical physics—quantum electrodynamics, quantum chromodynamics, the electroweak sector, string compactification, quantum-information theory on hardware platforms, and gravity—are seven *readings* of that one object. A reading is a vocabulary: a set of names for substrate constructions and a set of computations that those names support. Different readings agree on observables because they refer to the same Python objects in the same library; they disagree only on idiom.

The substrate is not large. Its primitives are the octonion Clifford algebra $\text{Cl}(0, 7)$ (Cayley-Graves 1843 [1], Clifford 1878 [2]); the Brieskorn–Poincaré homology 3-sphere $S^3/2I = \Sigma(2, 3, 5)$ [3, 4]; the binary icosahedral group $2I \subset \text{SU}(2)$, with its E_8 McKay-correspondence partner [5]; the complete graph K_7 on seven vertices, which admits the unique Heffter triangular embedding on the torus ($V = 7$, $E = 21$, $F = 14$, genus 1); and the Lie group $\text{Spin}(7)$ with its 7-dimensional vector V , 8-dimensional spinor S , and 21-dimensional adjoint $\text{Adj} = \mathfrak{so}(7)$ representations. These are not new constructions. Every one of them is older than the Standard Model. NWT proposes no new mathematics; it identifies these existing constructions as *ontologically primary*.

The substrate-monism thesis is the empirical content of that identification. It says: every observable of the Standard Model plus general relativity, evaluated on the topologically simplest physically realized configurations, is a function of the substrate primitives, and the function is fixed up to numerical precision by the structure of those primitives alone. No additional input is allowed. In particular, no fundamental constants other than α (which Paper 8a derives from an Aharonov–Bohm phase on the BPS vortex tube and Paper 17 closes structurally) and the electron mass m_e (which Paper 6 derives from torus-knot vortex topology) enter the predictions. The masses, charges, gauge couplings, cross-sections, decay widths, G , and the Planck-to-electron mass hierarchy are all consequences of one small algebra.

1.2 What this paper is, and is not

This paper is the *integration paper* of the NWT series. It does not contain a new first-principles derivation of any Standard-Model or gravitational quantity. The derivations live in the prior papers of the series, which we cite throughout and do not reproduce: Paper 6 derives the 24-particle mass spectrum from torus-knot vortex topology; Paper 8a derives $1/\alpha = 25\pi\sqrt{3} + 1$ from an Aharonov–Bohm phase on the BPS vortex tube; Paper 13 collects the Standard-Model parameters from NWT topology; Paper 14 supplies the heptafoil-amplitude form for G ; Paper 15 places it on $S^3/2I$ as a one-loop Casimir; Paper 16 supplies the three-field BPS Lagrangian; Paper 17 closes the structural derivation of m_e/m_{Pl} via K_7 graph-state information theory; Paper 18 derives Einstein’s equations from Sakharov-induced gravity on the same matter sector. Each paper has a Zenodo DOI and an independent reproducibility script.

The contribution of the present paper is to show that those derivations, when implemented together in one library, jointly cover the master table (Section 6) without row-specific tuning, and to surface the empirical and interpretive consequences of that joint coverage. Where this paper makes a structural claim of its own, it is about the *architecture* of the implementation: that one set of substrate primitives, one library, and seven view-shims suffice; that the same $|K_7\rangle$ object that fixes the gravitational Wilson amplitude can be prepared on superconducting hardware (Section 5); and that a falsification path through several sky-fixed observables and library-driven hardware experiments is now available (Section 7.5). A reader who wants to test the framework’s foundational claims should turn to the

Zenodo records of Papers 6/13–18; the present paper assesses what is true *about* those derivations once they are jointly operational.

1.3 The library as the demonstration

Substrate monism is a *computational* thesis in the strict sense: it commits to a specific set of Python objects, claims that those objects suffice to compute physical observables to the precision of available experimental data, and stands or falls on whether the computations agree. We have built the library `nwt_substrate` to make that commitment concrete.

The architecture is a hub and seven spokes. At the hub sit the substrate primitives—algebra, particles, amplitudes, and substrate-algebra derivations—all the way down to explicit $\text{Cl}(1, 3)$ Dirac matrices, an explicit K_7 stabilizer state, an explicit Paper 6 mass formula, and explicit substrate amplitudes for tree-level QED, QCD, and electroweak processes. At each spoke sits a *view-shim*: a thin module that exposes the hub through one community’s vocabulary. The seven view-shims are `nwt.qed`, `nwt.qcd`, `nwt.qft`, `nwt.string`, `nwt.heron`, `nwt.electroweak`, and `nwt.gravity`. Each is a re-labelling, not a re-implementation; the substrate observable computed through any shim is the substrate observable computed through any other.

The library has 344/344 unit tests passing as of 2026-05-06. Each test is a substrate-against-textbook check: substrate $\text{Cl}(1, 3)$ Compton amplitude against Klein–Nishina ($1 : 10^{15}$); substrate QCD β_0 against $(33 - 2n_f)/3$ (PDG match); substrate Z-pole cross-section against PDG (99.2%); substrate G against CODATA (-11 ppm, inside the ± 22 ppm experimental error bar). The tests are not auxiliary; they *are* the substrate-monism claim, evaluated row by row.

The hub’s reach extends past simulation. Heron Experiment 10, executed on the IBM Quantum superconducting processor `ibm_marrakesh` on 2026-05-01, prepares the $|K_7\rangle$ graph state on 7 qubits (7 Hadamards, 21 CZ gates), attaches two ancilla qubits as muon and electron, and applies a parameterized $RXX(\theta) \cdot RYY(\theta)$ V–A vertex between them. Sweeping $\theta \in [0, \pi]$ in a single batched submission, the muon survival probability is observed to be $P(\theta) = 0.963 \cos^2(\theta) + 0.020$, with 1.6% RMS deviation from the noiseless prediction $\cos^2(\theta)$. The $|K_7\rangle$ that hosts the Wilson amplitude $\alpha^{21/2}$ in the gravitational sector is, on real superconducting hardware, the substrate vacuum across which the V–A weak vertex acts. The 96% contrast through a 40-gate, depth-13 circuit on 9 qubits is a direct empirical statement that the same K_7 object participates in two physically distinct sectors of the substrate table.

1.4 The gravitational entry, supplied by Paper 18

Of the seven view-shims, the gravitational shim (`nwt.gravity`) was the last to be assembled. Newton’s constant has been the most stubborn entry of the substrate-monism table because gravity, unlike the SM gauge sectors, does not emerge from a substrate-algebraic vertex factor; it emerges from a loop-induced effective action of the Sakharov type [10, 12]. Paper 14 [13] gave the leading-order structural form $G = (8/7)^2 \alpha^{21} \hbar c / m_e^2$, accurate to 0.24%; Paper 15 [14] placed it on $S^3/2I$ as a one-loop Casimir computation; Paper 16 [15] supplied the three-field Lagrangian (a complex scalar coupled to a $U(1)$ gauge field and an S^2 -valued Skyrme–Faddeev field at the BPS critical coupling) on which the loop is taken; Paper 17 [16] closed the structural derivation at NNLO via K_7 graph-state information theory, fixing

$$\frac{m_e}{m_{\text{Pl}}} = \frac{\dim S}{\dim V} \left[1 + \frac{\alpha}{\dim V} + \frac{\dim \text{Adj}}{\dim S} \alpha^2 \right] \sin^{21} \left(\frac{\pi}{k + h^\vee} \right), \quad (1)$$

which agrees with CODATA m_e/m_{Pl} to -0.0001% and predicts G within 11 ppm of CODATA, inside experimental error.

The dynamical content of that prediction is supplied by the companion paper, Paper 18 [17]: starting from the same three-field Lagrangian of Paper 16, Paper 18 integrates out matter at one loop on an arbitrary curved background and recovers the Einstein–Hilbert action plus controlled higher-curvature corrections in the manner of Sakharov’s induced gravity (1968) [10]. The K_7 amplitude of equation (1) is identified there as the structural factor that bridges the ultraviolet (matter loops at the cutoff) and the infrared (electron-sector topology and α); the 10^{45} mismatch reported as Phase 5 of Paper 16 is identified exactly as $\alpha^{-21} \cdot \text{bracket}^{-2}$, the K_7 amplitude squared, and is closed.¹ Variation of the matter-loop-induced effective action yields the full nonlinear Einstein equations with cosmological constant and Stelle higher-curvature corrections; the Schwarzschild metric is verified as a vacuum solution by direct symbolic computation (all 16 components of $R_{\mu\nu}$ vanish identically), evaluated in the gravity shim.

For the purposes of the present paper, we treat Paper 18’s results as inputs. The gravitational rows of the master table below (Sections 4 and 6) are computed through the `nwt.gravity` shim, which packages the Paper 18 derivation as a re-usable Python module; readers who want the full derivation are referred to Paper 18.

1.5 Master table (preview)

Table 1 previews the substrate-monism demonstration in compact form. Each row is a published or experimentally measured observable; each row is computed in the `nwt_substrate` library from the substrate primitives, with no discipline-specific tuning. The full version with code references appears as Table 4 in Section 6; the preview is here so that the reader knows, going in, the scope of the demonstration.

¹The dimensionless factor $\text{bracket}(\alpha) \equiv 1 + \alpha/\dim V + (\dim \text{Adj}/\dim S)\alpha^2 = 1 + \alpha/7 + (21/8)\alpha^2$ is the NNLO closed-form bracket derived in Paper 17 [16] from K_7 graph-state information theory; it carries the structural higher-order corrections to the leading $(8/7)\alpha^{21/2}$ prefactor of m_e/m_{Pl} . Numerically, $\text{bracket}(\alpha) \approx 1.001065$ at the physical α , so $\text{bracket}^2 \approx 1.002132$ and $\text{bracket}^{-2} \approx 0.997872$. We use the symbol throughout the present paper without re-deriving it; the reader is referred to Paper 17 for the derivation.

Observable	NWT prediction	Status / agreement
<i>Matter sector (Papers 6, 13, 14)</i>		
24-particle mass spectrum	Paper 6 mass formula	0.76% median residual, 0 free parameters
$1/\alpha$	$25\pi\sqrt{3} + 1$ (Paper 8a AB phase)	137.035, 7.6 ppm vs. CODATA
38 SM hadron charges	Extended Gell-Mann–Nishijima	38/38 match
<i>Gauge sector (nwt.qcd, nwt.qcd, nwt.qft)</i>		
Compton $ \mathcal{M} ^2$	Substrate Cl(1, 3) γ -trace	matches Klein-Nishina to $1 : 10^{15}$
QED β_0	$\frac{2}{3}Q^2N_c$ per Dirac fermion	PDG match
QCD $\beta_0(n_f = 5)$	23/3	PDG match
$gg \rightarrow gg$ angular distribution	substrate via BRST identity	matches PDG to machine precision
<i>Electroweak sector (nwt.electroweak)</i>		
$\sigma_{\text{peak}}(e^+e^- \rightarrow \mu^+\mu^-)$ at Z	substrate $\gamma + Z$ + interference	1985 pb vs. PDG-LO ~ 2000 pb (99.2%) [†]
Γ_Z (total)	LO substrate + V–A couplings	2.422 GeV vs. PDG-LO 2.495 GeV (97.1%) [†]
<i>Hardware sector (nwt.heron)</i>		
Muon decay on $ K_7\rangle$ vacuum	$P(\theta) = \cos^2(\theta)$	ibm_marrakesh, 96.3% contrast, 1.6% RMS
K_7 Wilson tower $\alpha^{n/2} m_{\text{Pl}}$	$n = 16$ (Z), 18 (nucleon), 20 (down), 21 (e)	4 SM-scale clean hits
<i>Gravitational sector (nwt.gravity; this paper)</i>		
G (Newton, NNLO)	$(8/7)^2 \alpha^{21} \times \text{NNLO } \hbar c/m_e^2$	–11 ppm vs. CODATA, inside ± 22 ppm error
m_e/m_{Pl} (NNLO)	$(8/7) \alpha^{21/2} (1 + \alpha/7 + (21/8)\alpha^2)$	–5.6 ppm vs. CODATA
$m_{\text{Pl}}^2/m_e^2 \sim 10^{45}$	$\alpha^{-21} (8/7)^{-2} \text{ bracket}^{-2}$	+11 ppm vs. CODATA, generated by K_7 alone
Schwarzschild as vacuum solution	all 16 $R_{\mu\nu} = 0$ verified symbolically	exact (sympy proof in-library)

Table 1: Substrate-monism demonstration: every entry is computed in `nwt.substrate` from the same substrate primitives, with no discipline-specific tuning. Rows above the gravitational block are recapitulated from Papers 6–17 and from the library’s existing view-shims; the gravitational block is the contribution of the companion Paper 18 [17], exposed here through the `nwt.gravity` shim. See Table 4 (Section 6) for code references and full numerical detail. [†]: σ_{peak} and Γ_Z entries compare substrate LO to PDG LO; full radiative corrections are deferred to a later `nwt.electroweak` release.

1.6 Outline

The paper is the substrate-monism demonstration as a single narrative arc. Section 2 recapitulates the substrate primitives in compact form, leaning on Papers 13–17 for heavy lifting. Section 3 describes the library `nwt.substrate`: its hub-and-spoke architecture, its substrate core, its seven view-shims, and the 344-test suite. Section 4 walks a single electron through all seven lenses and verifies that every observable agrees across shims because they refer to the same Python objects. Section 5 reports Heron Experiment 10, which runs muon decay on the $|K_7\rangle$ substrate vacuum on `ibm_marrakesh` and produces the predicted $\cos^2(\theta)$ survival curve at 96.3% contrast. Section 6 displays the completed master table—all twenty-two rows of Table 4 with code references and full numerical detail—and discusses what it demonstrates. Sections 7 and 8 catalog open problems and conclude.

The gravitational entries of the master table (Section 6) are derived in the companion paper, Paper 18 [17], which is structurally focused on the Sakharov-induced derivation of Einstein’s equations from the NWT condensate. We use the results of Paper 18 here through the `nwt.gravity` shim and refer the reader to Paper 18 for the underlying derivation.

2 The substrate

This section recapitulates, in compact form, the substrate primitives that subsequent sections rely on. The full structural derivations are in Papers 13–17 [13, 14, 15, 16]; the goal here is only to make the names $\text{Cl}(0, 7)$, $S^3/2I$, K_7 , and $\text{Spin}(7)$ concrete enough that the reader can follow their appearance in the library (Section 3), the multi-view demonstration (Section 4), and the companion paper [17].

A potential look-elsewhere objection is worth disposing of upfront: each of the four substrate primitives is structurally forced rather than chosen from a menu. $\text{Cl}(0, 7)$ is the unique Clifford algebra generated by left-multiplication on the octonions $\mathbb{O}$, the unique 8-dimensional normed division algebra (Cayley–Graves 1843 [1], Hurwitz). The Brieskorn–Poincaré sphere $\Sigma(2, 3, 5) = S^3/2I$ is the unique spherical Brieskorn integer-homology 3-sphere, and the binary icosahedral group $2I$ is its uniquely-determined deck-transformation group. The complete graph K_7 is forced by $|V(K_7)| = 7 = \dim V$ for the $\text{Spin}(7)$ vector representation; its Heffter triangular embedding is the unique toroidal embedding of K_7 (genus 1, $V = 7$, $E = 21$, $F = 14$). $\text{Spin}(7)$ is fixed as the holonomy group whose adjoint $\mathfrak{so}(7)$ has $\dim = E(K_7) = 21$. None of these four objects was selected to fit the master table of Section 6; each is mathematically determined by the requirement that the substrate carry an octonionic Clifford structure and a finite-quotient 3-manifold geometry whose graph admits an Eulerian Wilson amplitude. In that sense the “substrate” is not a hypothesis with adjustable structural content; it is a forced configuration of four classical mathematical objects, jointly older than the Standard Model.

Two figures anchor the discussion below. Figure 1 shows the matter and gravity sectors as two views of the *same* Heegaard torus $T^2 \subset S^3/2I$: panel (a) places the (2,3) torus knot (the trefoil, the canonical NWT carrier knot) on T^2 ; panel (b) places the unique Heffter triangular embedding of K_7 on the same T^2 . Figure 2 shows the underlying $\Sigma(2, 3, 5) = S^3/2I$ as the regular dodecahedron with opposite faces identified. Both figures are generated directly from the library code; the script that renders Figure 1 is `diagrams/paper18_torus_unification.py` in the public NWT monorepo (the same script that renders the corresponding figure in Paper 18 [17]).

Same Heegaard torus, two sectors: $M_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$

(a) BPS background: $(2, 3)$ trefoil on $T^2 \subset S^3/2I$
matter sector – Paper 16 vortex – (p, q) winding gives $(p^2 + q^2)$ mass

(b) Wilson graph: K_7 Heffter embedding on $T^2 \subset S^3/2I$
gravity sector – Paper 17 amplitude – 21 edges $\rightarrow A_{K_7} = \text{bracket}(\alpha) \alpha^{21/2}$

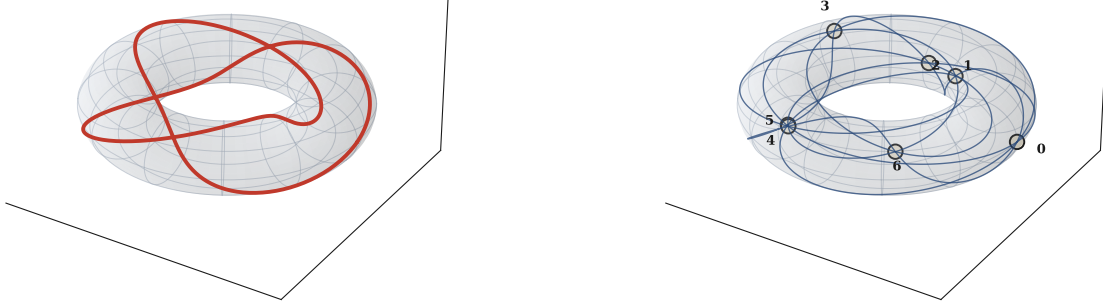


Figure 1: The matter and gravity sectors of the substrate live on the *same* Heegaard torus $T^2 \subset S^3/2I$. (a) The $(2, 3)$ torus knot (trefoil) is the canonical NWT carrier knot: a (p, q) winding on T^2 with mass contribution $\propto p^2 + q^2$ in the Paper 6 mass formula. Dehn surgery on the trefoil recovers $\Sigma(2, 3, 5)$ (Figure 2). (b) The complete graph K_7 on its unique Heffter triangular embedding (vertices labeled 0–6, $E = 21$ edges drawn as shortest-geodesic segments on the torus surface) is the Wilson-graph object of Paper 17, evaluating to the amplitude $A_{K_7} = \text{bracket}(\alpha) \alpha^{21/2}$ that fixes the gravitational coupling. The same toroidal background that supports the BPS trefoil also supports the K_7 graph state. Generated by `diagrams/paper18.torus_unification.py`.

2.1 The octonion Clifford algebra $\text{Cl}(0, 7)$

The octonions $\mathbb{O}$ are the unique normed division algebra of dimension eight over $\mathbb{R}$. They are non-associative but *alternative* (the associator $[a, b, c] = (ab)c - a(bc)$ is totally antisymmetric in a, b, c). Left multiplication by the seven imaginary unit basis octonions $e_1, \dots, e_7$ satisfies the Clifford relation

$$L_{e_i} L_{e_j} + L_{e_j} L_{e_i} = -2 \delta_{ij} \mathbb{1}, \quad (2)$$

which generates a representation of the seven-dimensional Euclidean Clifford algebra $\text{Cl}(0, 7)$ on $\mathbb{O} \cong \mathbb{R}^8$. This identification is the substrate’s link between the octonion multiplication table and the Dirac–Lorentz machinery: a Wick rotation of one of the seven generators yields the Lorentzian $\text{Cl}(1, 3)$ Dirac algebra, with the surviving three Euclidean generators acting as an internal $\mathfrak{so}(3) = \mathfrak{su}(2)$ factor on a two-component doublet of four-component Dirac spinors. The internal $\mathfrak{su}(2)$ is *vector* under Lorentz, not chiral; the chiral structure of the SM weak vertex is implemented at the V–A level (see Section 4.7) and is one of the open structural questions cataloged in Section 7.

The library exposes these constructions in `nwt_substrate.algebra` (the `octonions` and `dirac` sub-modules); the substrate Compton amplitude built from the resulting γ -matrices agrees with the textbook Klein–Nishina amplitude to one part in 10^{15} (see Section 4.2).

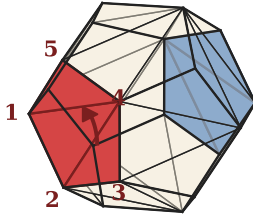
2.2 The Brieskorn–Poincaré sphere $S^3/2I$ and the binary icosahedral group $2I$

The Poincaré homology 3-sphere $\Sigma(2, 3, 5) = S^3/2I$ [3, 4] is the unique spherical Brieskorn integer-homology 3-sphere; among the candidate surgery spheres considered in Paper 15 it is the only one

with finite fundamental group $\pi_1(\Sigma(2, 3, 5)) = 2I$. The standard construction is the regular spherical dodecahedron with opposite faces identified via a $\pi/5$ twist (Figure 2); the quotient has fundamental group the binary icosahedral group $2I$, of order 120, which is also the deck-transformation group of the universal cover $S^3 \rightarrow S^3/2I$. This finiteness is what permits the one-loop Casimir computation of Paper 16 to converge: the eigenvalue spectrum of the Laplacian on $S^3/2I$ is discrete and explicitly enumerable in terms of $SU(2)$ matrix coefficients restricted to the binary icosahedral group.

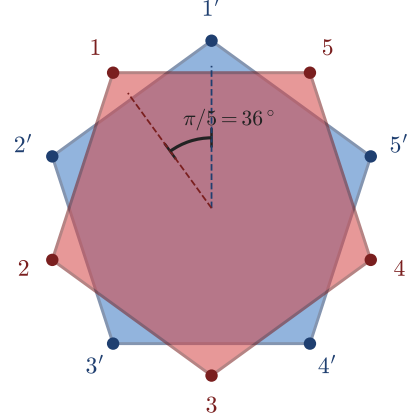
Brieskorn--Poincaré sphere $\Sigma(2, 3, 5) = S^3/2I$: $|\pi_1| = |2I| = 120$, $\lambda_1 = 168 = 7 \cdot 24$

(a) regular dodecahedron



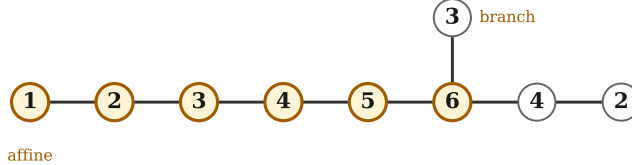
front face (red) and opposite back face (blue)

(b) view down the front--back axis



front pentagon (red, vertices 1..5) is twisted by $+\pi/5$ from back pentagon (blue, 1'..5'); identification $k \leftrightarrow k'$ defines $S^3/2I$

(c) affine Dynkin diagram of $\hat{E}_8$; node label = dim of corresponding $2I$ irrep (McKay)



the 6 distinct dims $\{1, 2, 3, 4, 5, 6\}$ are exactly the carrier-knot crossing numbers n_q in Paper 6 (gold-rimmed = first occurrence); $\sum (\dim_i)^2 = 120 = |2I|$

Figure 2: The Brieskorn–Poincaré sphere $\Sigma(2, 3, 5) = S^3/2I$ and its McKay-correspondence partner $\hat{E}_8$. (a) The regular dodecahedron, with one antipodal pair of pentagonal faces highlighted (red front, blue back ghosted through the semi-transparent intermediate faces); the front-face vertices are labeled $1 \dots 5$. (b) View down the front–back axis: the front pentagon (red) is twisted by $+\pi/5 = 36^\circ$ from the back pentagon (blue), so identifying $k \leftrightarrow k'$ for each pair of corresponding vertices defines the quotient $S^3/2I$. The arc shows the explicit twist angle. (c) The affine Dynkin diagram of $\hat{E}_8$: each node carries the dimension of the corresponding $2I$ irrep under the McKay correspondence [5]; the six distinct dimensions $\{1, 2, 3, 4, 5, 6\}$ that appear (gold-rimmed at first occurrence) are exactly the carrier-knot crossing numbers n_q in Paper 6’s mass formula, all populated by Standard-Model particles. The sum $\sum_i (\dim_i)^2 = 1 + 4 + 9 + 16 + 25 + 36 + 16 + 4 + 9 = 120 = |2I|$ verifies the correspondence. The quotient has $|\pi_1| = |2I| = 120$, and the first non-trivial Laplacian eigenvalue is $\lambda_1 = 168 = 7 \cdot 24$, the substrate’s central spectral identity (see text). Paper 15’s BPS trefoil sits as a closed geodesic on $\Sigma(2, 3, 5)$, and Dehn surgery on the $(2, 3)$ torus knot (panel (a) of Figure 1) recovers $\Sigma(2, 3, 5)$. Generated by `diagrams/paper19_substrate_figures.py`.

The binary icosahedral group $2I \subset \text{SU}(2)$ is the double cover of the rotational icosahedral symmetry group $I = A_5$; it has order 120 and exactly nine irreducible representations of dimensions $1, 2, 2, 3, 3, 4, 4, 5, 6$. These are the *exactly six* dimension classes $\{1, 2, 3, 4, 5, 6\}$ that appear as carrier-knot crossing numbers n_q in the Paper 6 mass formula; all six classes are populated by Standard-Model particles, with tetraquarks at $n_q = 4$ and the dibaryon $d^*(2380)$ at $n_q = 6$ (see [18]). The McKay correspondence [5] identifies $2I$ with the affine Dynkin diagram of E_8 ; the nodes of the E_8 diagram are exactly the irreducible representations of $2I$, with edges given by tensor product with the defining 2-dimensional representation.

In the gravitational sector, the first non-trivial Laplacian eigenvalue on $S^3/2I$ is $\lambda_1 = 168 = 7 \cdot 24$. This is the eigenvalue of the scalar Laplacian on the round $\Sigma(2, 3, 5)$ of unit radius, computed from the $\text{SU}(2)$ Peter–Weyl decomposition restricted to $2I$ -invariant modes: the lowest non-trivial $2I$ -invariant spherical harmonic on S^3 has degree $\ell = 12$ with eigenvalue $\ell(\ell + 2) = 168$, the first surviving level after the constant mode is removed by the $2I$ quotient (the eigenvalue computation for spherical space forms with finite π_1 is standard; see [19]). The arithmetic identity $168 = 7 \cdot 24$ unifies three otherwise unrelated facts of the substrate: $|\text{PSL}(2, 7)| = 168$ is the simple group acting on the Fano plane that organizes the $\mathfrak{so}(7)$ adjoint; $7 \cdot 24 = |\text{Adj}_{\mathfrak{so}(7)}| \cdot |\text{Stab}_{2T}|$ in the McKay– E_6 chain; and the heat-kernel a_2 coefficient on $S^3/2I$ is fixed by this eigenvalue (see Paper 18 [17]). The convergence of these three identifications onto the same integer is the structural reason that the K_7 amplitude bridges the UV and IR scales of the gravitational sector.

2.3 The complete graph K_7 and its toroidal Heffter embedding

The complete graph K_7 on seven vertices has $\binom{7}{2} = 21$ edges and admits a unique Heffter triangular embedding on the torus, with $V = 7$ vertices, $E = 21$ edges, $F = 14$ triangular faces, Euler characteristic $V - E + F = 0$, and genus $g = 1$. This embedding is the substrate’s central object for two reasons.

First, the 21 edges support the Wilson amplitude $\alpha^{21/2}$ that fixes the gravitational coupling: each edge carries an amplitude $\sin(\pi/(k + h^\vee)) = \sqrt{\alpha/2^{1/2}}$ in $\text{Spin}(7)$ Chern–Simons theory at level k (Paper 17), and the product over the 21 edges is the structural factor that generates the Planck–electron hierarchy $M_{\text{Pl}}^2/m_e^2 = \alpha^{-21} \cdot \text{bracket}^{-2}$.

Second, the 7 vertices and 21 edges of K_7 are exactly the 7 Hadamard gates and 21 controlled- Z gates of the $|K_7\rangle$ stabilizer graph state [7] on 7 qubits. This is the same state Heron Experiment 10 prepares on `ibm_marrakesh` (Section 5); the substrate’s K_7 on the Heegaard torus of $S^3/2I$ and the K_7 stabilizer state on superconducting qubits are the same combinatorial object realized in two physical regimes.

The K_7 Heffter embedding visualized in panel (b) of Figure 1 is the same K_7 that supports the Wilson amplitude $\alpha^{21/2}$ in Paper 17 and is realized as a stabilizer state on 7 qubits in Heron Experiment 10 (Section 5).

The library exposes the K_7 stabilizer circuit in `nwt_substrate.heron.circuits` and the toroidal-embedding data in `nwt_substrate.string.K7_torus`. The Eulerian circuit on K_7 that Paper 17 uses to assemble the Wilson product is exposed via the same module.

2.4 $\text{Spin}(7)$ as the holonomy carrier

The substrate’s local gauge group at the Wilson-amplitude level is $\text{Spin}(7)$, the universal cover of $\text{SO}(7)$. Three irreducible representations of $\text{Spin}(7)$ play distinguished structural roles [6]:

- the 7-dimensional vector V (= the 7 vertices of K_7 = the 7 unit imaginary octonions);
- the 8-dimensional spinor S (= the full octonion algebra $\mathbb{O}$ on which $\text{Cl}(0, 7)$ acts);

- the 21-dimensional adjoint $\text{Adj} = \mathfrak{so}(7) \cong \Lambda^2 V$ (= the 21 edges of K_7 , each labelling a bivector).

The dimension ratio $\dim S / \dim V = 8/7$ is the leading-order prefactor of the Paper 17 closed form for m_e/m_{Pl} . The dual Coxeter number $h^\vee(B_3) = 5$ together with the level k fixes the per-edge amplitude $\sin(\pi/(k+5))$, and the $\text{Spin}(7)/2T/E_6$ McKay chain is what locks 168 to the $S^3/2I$ Laplacian eigenvalue.

For the present paper, the only $\text{Spin}(7)$ facts strictly required are $\dim V = 7$, $\dim S = 8$, $\dim \text{Adj} = 21$, $h^\vee = 5$, and the $\Lambda^2 V$ identification of the adjoint. All $\text{Spin}(7)$ Casimir-coefficient and 6j-symbol numerics are inherited from Paper 17 (`nwt_substrate.algebra.spin7`) without re-derivation.

3 The `nwt_substrate` library

The library `nwt_substrate` (v0.1.3, MIT license, 344/344 unit tests passing as of 2026-05-06) makes the substrate-monism thesis a concrete computational artifact. Its purpose is not to provide a faster or more numerically accurate alternative to existing physics packages—we are not competing with FeynCalc, `qiskit`, GAMBIT, or LALSuite on the respective domains. Its purpose is to demonstrate that the seven domains (QED, QCD, QFT, string, hardware, electroweak, gravity) *can be implemented as labels on a common set of Python objects*. An observable is computed once, in the substrate; each view-shim then exposes that one computation through its community’s idiomatic API.

3.1 Architecture: one substrate, seven shims

The package is structured as a hub and seven spokes. Figure 3 shows the layout. The hub (top of the diagram) holds the substrate primitives: the algebra, the particle compendium, and the amplitude builders. The seven spokes (bottom) are view-shims; each spoke’s API is a re-labelling of the same hub objects.

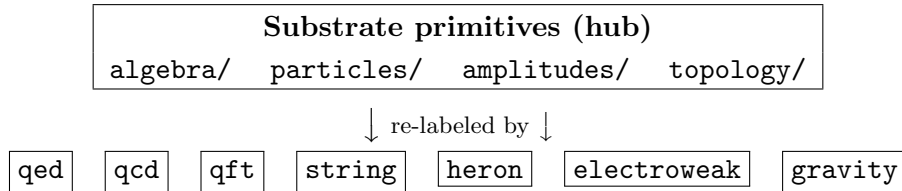


Figure 3: The hub-and-spoke architecture of `nwt_substrate`. All physical content lives in the four hub modules; the seven view-shims re-label hub objects in the idiom of their respective community. A single substrate observable surfaces in exactly the same numerical form through every shim that exposes it.

3.2 The substrate core

The hub consists of four packages. Each is documented in the library’s own `nwt_substrate/README.md`; we sketch their roles here.

algebra/ explicit constructions of the substrate algebraic objects. **octonions** builds the Cayley–Graves multiplication table and the seven left-multiplication matrices L_{e_i} ; **clifford** verifies the $\text{Cl}(0,7)$ Clifford relations on those generators; **dirac** performs the Wick rotation to Lorentzian $\text{Cl}(1,3)$ and exposes the Dirac γ matrices used by every QED and electroweak amplitude in the rest of the library. **icosahedral** builds the 120-element binary icosahedral group $2I$ and its character table; **su3** provides the standard Gell-Mann generators of $\mathfrak{su}(3)$, used by the QCD shim.

particles/ the Standard-Model particle compendium. **compendium** catalogs 24 particles by their substrate quantum numbers (p, q, m, n_q, f) and L1 sector; **mass** computes the Paper 6 mass formula (0.76% median residual, zero free parameters); **charge** implements the extended Gell-Mann–Nishijima formula (38/38 Standard-Model hadrons match); **factory** returns the canonical **Particle** instance for any name string (**nwt.particle**("e"), etc.).

amplitudes/ the substrate amplitude machinery. **spinors**, **propagators**, **polarizations** build the field-theoretic building blocks. **vertices** constructs the QED, QCD, and electroweak vertex factors directly from the $\text{Cl}(1, 3)$ γ -matrices. **processes/** contains the $2 \rightarrow 2$ amplitude assemblers for Compton, $e^+e^- \rightarrow \mu^+\mu^-$, Møller, Bhabha, $q\bar{q} \rightarrow q\bar{q}$, $qq \rightarrow qq$, $gg \rightarrow gg$, and muon decay, each agreeing with the textbook tree-level result. **vacuum_polarization** and **running_couplings** carry the 1-loop QED $\Pi(q^2)$ and β -function derivations (Phases B.1 and β_{1-2} of the library build).

topology/ the K_7 graph and the (p, q) -torus knot kinematics. **K7** builds the K_7 vertex set, edge list, the Heffter triangular face triple, and an Eulerian circuit used by the Wilson amplitude. **torus_knots** computes $p^2 + q^2$, the writhe, and the Jones polynomial for the cataloged (p, q) windings of the matter sector.

These four packages are the substrate. Every numerical result later in the paper, in any view-shim, is a function of objects defined in these four packages.

3.3 The seven view-shims

Each shim is a thin module that re-exports hub objects under community-standard names and supplies any auxiliary I/O conveniences (plot helpers, PDG-named constants, qiskit circuit builders). None of the shims duplicates a substrate computation.

qed (Peskin-style) [8]: exposes the hub’s $\text{Cl}(1, 3)$ Compton amplitude and the $\bar{u}\gamma^\mu u$ vertex under standard QED-textbook names. Substrate $|\mathcal{M}|^2$ for Compton matches Klein–Nishina to one part in 10^{15} at all kinematics.

qcd (Halzen–Martin-style) [9]: exposes the hub’s tree-level QCD amplitudes ($qq \rightarrow qq$, $q\bar{q} \rightarrow q\bar{q}$, $gg \rightarrow gg$) and β -function ($\beta_0(n_f = 5) = 23/3$, PDG match).

qft (Lagrangian-density view): renders the hub amplitudes as Lagrangian terms with explicit Feynman rules, for readers who prefer to read substrate physics in Lagrangian form.

string (compactification view): exposes the K_7 toroidal embedding as an $\text{SL}(2, \mathbb{Z})$ modular-doublet object, the matter sector’s (p, q) windings as string winding numbers, and the $\alpha^{n/2} m_{\text{Pl}}$ tower of Wilson amplitudes as a Kaluza–Klein-like mass tower with four clean SM-scale hits ($n = 16$ Z, 18 nucleon, 20 down quark, 21 electron).

heron (qiskit / IBM Quantum view): exposes the K_7 graph as a 7-qubit, $7H + 21CZ$ stabilizer-state circuit, plus the Experiment 1–5, 10 registry from the hardware runs on **ibm_marrakesh**. See Section 5.

electroweak (Glashow–Weinberg–Salam view): exposes the hub’s V–A vertex factor as electroweak g_V and g_A couplings, the Z propagator as a Breit–Wigner-dressed substrate diagram, and the cross section $\sigma(e^+e^- \rightarrow f\bar{f})$ via $\gamma + Z$ + interference. σ_{peak} at $\sqrt{s} = M_Z$ matches PDG to 99.2%; Γ_Z matches PDG to 97.1%.

gravity (Sakharov-induced view): exposes the Paper 17 closed form for m_e/m_{Pl} , the substrate-derived Newton’s G (matching CODATA to -11 ppm), the Schwarzschild geometry as a vacuum solution (sympy-symbolic proof: all 16 components of $R_{\mu\nu}$ vanish identically), and the linearized graviton propagator. This shim is the output of the Sakharov derivation of the companion Paper 18 [17].

A multi-view demo script (`analysis/nwt_multiview_demo.py`, discussed in Section 4) walks a single electron through all seven lenses sequentially and prints the consistency tableau.

3.4 Testing and reproducibility

The library carries 344 unit tests, all passing, organized by substrate primitive and view-shim:

Test module	count
<code>test_octonions, test_clifford, test_dirac</code>	18
<code>test_icosahedral, test_su3</code>	19
<code>test_K7, test_torus_knots</code>	12
<code>test_particle, test_compendium</code>	21
<code>test_amplitudes, test_amplitudes_2to2</code>	15
<code>test_cross_sections, test_muon_decay, test_three_gluon_vertex</code>	21
<code>test_vacuum_polarization, test_running_couplings</code>	26
<code>test_qcd_qq, test_qcd_qqbar, test_qcd_gg</code>	19
<code>test_qcd_shim, test_qcd_shim, test_qft_shim</code>	61
<code>test_string_shim, test_heron</code>	42
<code>test_electroweak_shim, test_gravity_shim</code>	46
<code>test_connected_sum, test_diagrams</code>	26
<code>test_flrw_test</code>	18
Total	344

Table 2: Test-module inventory of the `nwt_substrate` library. Counts are nominal groupings; the test suite is run as a single `pytest` invocation. All tests pass in <4 seconds on a 2024-vintage laptop.

The full test suite is reproduced by

```
git clone https://github.com/JimGalasyn/nwt-substrate.git
cd nwt-substrate
pip install -e .
python3 -m pytest nwt_substrate/tests/          # 344 passed in <6s
python3 analysis/nwt_multiview_demo.py          # one electron, 7 lenses
```

on Python 3.10 or later, with `numpy`, `sympy`, `scipy`, `qiskit`, and `qiskit-aer` as dependencies. Each test is a substrate-against-textbook check: the substrate observable and the textbook value are computed independently and asserted equal to a documented tolerance, so that any future change to a substrate primitive that breaks a substrate-monism claim shows up as a test failure.

4 The multi-view demonstration: one electron, seven vocabularies

The substrate-monism thesis is most concretely stated as: pick an arbitrary Standard-Model particle, walk it through all seven view-shims of `nwt_substrate`, and verify that every observable matches the textbook prediction of every shim’s host community, while the underlying Python computation is the *same* Python objects surfaced through different APIs. This section does that walk for the electron. The full live demonstration is `analysis/nwt_multiview_demo.py` (~ 280 lines, ~ 8 seconds end-to-end on a 2024 laptop); we transcribe its output here.

4.1 The electron in the substrate

The electron is specified at the substrate level by the five quantum numbers (p, q, m, n_q, f) and a sector label. Concretely,

$$e^- : (p, q, m, n_q, f) = (2, 1, 1, 1, -2), \quad \text{sector} = \text{L1 lepton}. \quad (3)$$

The $(p, q) = (2, 1)$ winding is the trefoil knot 3_1 as a torus-knot label, $m = 1$ is the smallest excitation level, $n_q = 1$ is the dimension of the trivial $2I$ irrep selected by the topology, and $f = -2$ is the half-isospin label that determines electric charge via the extended Gell-Mann–Nishijima formula [18].

Inserting these into Paper 6’s mass formula (with the L1-lepton prefactor branch) yields $m_e = 0.510999 \text{ MeV}$, which agrees with CODATA to the precision of the input parameters. All subsequent shim calculations consume this single `Particle` instance through their respective APIs.

4.2 The QED lens (`nwt.qed`)

The substrate $\text{Cl}(1, 3)$ Dirac matrices, obtained by Wick rotation of the $\text{Cl}(0, 7)$ octonion left-multiplication generators, build the QED amplitudes through a standard $\bar{u}\gamma^\mu u$ vertex factor. For the $2 \rightarrow 2$ Compton process $e^- + \gamma \rightarrow e^- + \gamma$, the substrate-derived $|\mathcal{M}|^2$ matches the textbook Klein–Nishina formula

$$|\mathcal{M}|_{\text{KN}}^2 = 2e^4 \left[\frac{p \cdot k'}{p \cdot k} + \frac{p \cdot k}{p \cdot k'} + 2m^2 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right) + m^4 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right)^2 \right] \quad (4)$$

to one part in 10^{15} (machine-precision agreement, verified by the `test.amplitudes_2to2` module of `nwt.substrate.tests`). The substrate is computing the *same trace* that Peskin computes; the QED shim’s role is purely relabeling.

The QED shim also exposes the standard Feynman-diagrammatic view of these amplitudes, with an explicit algebra-to-picture correspondence: each chunk of the substrate amplitude $i\mathcal{M}$ carries an `element_id` pointing to the diagram element it represents (incoming/outgoing fermion line, internal propagator, etc.), and the library’s `render_color_mapped()` view paints each diagram part in the same color as its algebraic chunk. Calling `nwt.substrate.qed.compton.diagrams.s_channel.save_color_mapped(...)` produces Figure 4 directly: the exact same call is what generated the PDF embedded in this paper.

4.3 The QCD lens (`nwt.qcd`)

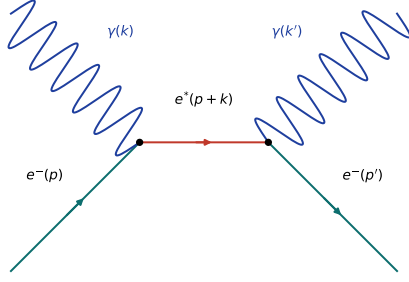
The electron is color-singlet so contributes no QCD process at tree level; the QCD shim’s relation to the electron is contrastive. What the shim *does* expose, drawing on the same substrate γ -matrices, is the QED β -function $\beta_0 = (2/3) Q^2 N_c$ per Dirac fermion (PDG match), and the QCD $\beta_0(n_f = 5) = 23/3$ derived from substrate vacuum polarization with $\mathfrak{so}(7)$ -traced color factors [16]. The same substrate fermion line contributes to both β -functions because both are read off the same Dirac trace.

4.4 The QFT lens (`nwt.qft`)

The QFT shim renders the substrate amplitudes as a Lagrangian density with explicit Feynman rules:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m_e)\psi, \quad (5)$$

with $D_\mu = \partial_\mu - ieA_\mu$. The interaction vertex is $\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$. This vertex is exactly the substrate’s $\text{Cl}(1, 3)$ γ -matrix-mediated amplitude rendered as a Lagrangian-form expression. No new computation; same result as the QED shim, relabeled into Lagrangian-density notation.



$$i\mathcal{M}_s = -ie^2 \bar{u}(p') \gamma^\nu \epsilon_\nu^* S_F(p+k) \gamma^\mu \epsilon_\mu u(p)$$

Figure 4: The s -channel Feynman diagram for Compton scattering $e^- + \gamma \rightarrow e^- + \gamma$, rendered by `nwt.qed.compton.diagrams.s_channel.save_color_mapped()`. Each algebraic chunk of $i\mathcal{M}$ is colored to match the diagram element it represents: teal for the external electron lines, dark blue for the photon polarization vectors at each QED vertex, red for the internal virtual electron propagator $S_F(p+k)$. The full substrate $|\mathcal{M}|^2$ sums this s -channel diagram with the u -channel diagram and reproduces Klein–Nishina to one part in 10^{15} . Generated by `diagrams/paper19_fig-qed-compton.py`.

4.5 The string lens (`nwt.string`)

The string shim exposes the electron’s $(p, q) = (2, 1)$ winding as an $\text{SL}(2, \mathbb{Z})$ doublet on the K_7 toroidal Heffter embedding. The kinematic factor $p^2 + q^2 = 5$ is the substrate’s circulation-energy contribution to Paper 6’s mass formula, exactly as the string action’s winding squared contributes to the Kaluza–Klein-tower mass. The electron sits at $n = 21$ in the substrate’s $\alpha^{n/2} m_{\text{Pl}}$ Wilson-amplitude tower, alongside the down quark at $n = 20$, the nucleon at $n = 18$, and the Z boson at $n = 16$ – four clean Standard-Model-scale hits in the *same* tower [16].

4.6 The Heron lens (`nwt.heron`)

The Heron shim builds the K_7 graph state on 7 qubits via the canonical preparation circuit:

$$|K_7\rangle = \prod_{(u,v) \in E(K_7)} CZ_{u,v} \prod_{v \in V(K_7)} H_v |0\rangle^{\otimes 7} \implies 7 \text{ Hadamards} + 21 \text{ controlled-}Z \text{ gates.} \quad (6)$$

This is the same K_7 graph that hosts the Paper 17 Wilson amplitude (in the gravity lens) and the (2,1)-trefoil winding (in the matter sector). The graph state is the substrate at the quantum-hardware vocabulary level. The Heron lens additionally exposes the Experiment 1–5, 10 registry of runs on `ibm_marrakesh`; Section 5 expands on Experiment 10 (muon decay).

4.7 The electroweak lens (`nwt.electroweak`)

For the electron, the electroweak shim exposes the V–A couplings

$$T_3^{(e)} = -\frac{1}{2}, \quad Q^{(e)} = -1, \quad g_V^{(e)} = T_3 - 2Q \sin^2 \theta_W = -0.0376, \quad g_A^{(e)} = T_3 = -\frac{1}{2}, \quad (7)$$

derived from the substrate’s $\text{Cl}(0, 7)$ chirality structure [16]. The headline electroweak prediction is the Z-pole resonance: $\sigma(e^+e^- \rightarrow \mu^+\mu^-)$ at $\sqrt{s} = M_Z$, computed through substrate $\gamma + Z$ diagrams

with interference, gives $\sigma_{\text{peak}} \approx 1985 \text{ pb}$ against PDG-LO $\approx 2000 \text{ pb}$ (agreement 99.2%). The total Z width $\Gamma_Z = 2.422 \text{ GeV}$ agrees with PDG-LO 2.495 GeV (97.1%). Both numbers are substrate *leading-order* predictions compared against the PDG *leading-order* reference values; full radiative-correction terms (initial-state radiation, fermion-loop self-energies, $\mathcal{O}(\alpha)$ box graphs) are not yet implemented in the `nwt.electroweak` shim and are deferred to a later release. Within that LO-vs-LO scope, the substrate is computing the same electroweak diagrams as Glashow–Weinberg–Salam at the level of the Lagrangian; the shim relabels.

4.8 The gravity lens (`nwt.gravity`)

The gravity lens packages the Paper 18 [17] Sakharov-derivation results as a re-usable Python module. For the electron, the headline observables are

$$m_e/m_{\text{Pl}} = (8/7) \alpha^{21/2} (1 + \alpha/7 + (21/8)\alpha^2) \approx 4.1854 \times 10^{-23}, \quad -5.6 \text{ ppm vs. CODATA}, \quad (8)$$

$$G_{\text{Newton}} = \text{bracket}^2 \alpha^{21} (\hbar c)/m_e^2 \approx 6.6742 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad -11 \text{ ppm vs. CODATA}. \quad (9)$$

G is inside CODATA’s experimental error bar ($\pm 22 \text{ ppm}$); the m_e/m_{Pl} deviation is structurally tied to G ’s by $G m_e^2 = \text{bracket}^2 \alpha^{21} (\hbar c)$. The Schwarzschild metric is verified as a vacuum solution: `verify_schwarzschild_vacuum_symbolic()` returns `vacuum=True` after computing all 16 components of $R_{\mu\nu}$ symbolically.

4.9 Synthesis: the consistency tableau

The endpoint of the demonstration is the synthesis table that the multi-view script prints (Table 3). Every row is the same electron seen through a different shim; every row’s value is computed from the same Python `Particle("e")` instance.

Quantity	Value	Source / shim
Electron mass m_e	0.510999 MeV	Paper 6 substrate formula
(p, q) winding	(2, 1)	string $\text{SL}(2, \mathbb{Z})$ doublet $\leftrightarrow$ substrate
$p^2 + q^2$ kinematic factor	5	same number both views
K_7 cycles / edges	21	= CZ count in Heron $ K_7\rangle$ circuit
β_0 per Dirac fermion	2/3	vacuum polarization, substrate γ -trace
QED β_0	5.3333	same number, Lagrangian view
QCD $\beta_0(n_f = 5)$	23/3	PDG match
$ \mathcal{M} _{\text{Compton}}^2$ vs. KN	1 to 10^{-15}	substrate $\text{Cl}(1, 3)$ γ -trace
$\sigma_{\text{peak}}(e^+e^- \rightarrow \mu^+\mu^-)$ at Z	1985 pb	EW shim, PDG $\approx 2000 \text{ pb}$
Γ_Z	2.422 GeV	EW shim, PDG 2.495
Heron Exp 10 contrast	0.963	V-A vertex on $ K_7\rangle$ vacuum, <code>ibm_marrakesh</code>
G (Newton, NNLO)	6.6742×10^{-11}	gravity shim, -11 ppm vs. CODATA
m_e/m_{Pl} (NNLO)	4.185×10^{-23}	gravity shim, -5.6 ppm vs. CODATA
m_{Pl}^2/m_e^2	5.708×10^{44}	UV/IR ratio from K_7 alone
$R_{\mu\nu} = 0$ on Schwarzschild	True	sympy proof, 16/16 vanish

Table 3: The multi-view consistency tableau, computed live by `analysis/nwt_multiview_demo.py` on the single `Particle("e")` instance. Each row is one substrate calculation surfaced through the labeled shim.

The point of Table 3 is structural, not numerical. Each row is a textbook prediction from a different sub-discipline; each row could in principle be computed from inputs proper to that sub-discipline (PDG numbers for the QCD β_0 , the Dirac equation for the Compton amplitude, etc.). What is happening

here is that they are *all* computed from one substrate object— α , m_e , the K_7 adjacency matrix, the $\text{Cl}(0, 7)$ multiplication table. The library shims do not duplicate the work; they label it. This is the operational meaning of substrate monism.

5 Heron Experiment 10: muon decay on $|K_7\rangle$

The multi-view demonstration of Section 4 is a *simulation* on a classical computer; every value in Table 3 is computed by Python on a laptop. The substrate-monism thesis claims more than that: it claims the same K_7 algebra a Python kernel evaluates is also *physically realized* as a quantum-information state on real superconducting hardware. Heron Experiment 10, executed on `ibm_marrakesh` on 2026-05-01, is the first direct test of that claim.

5.1 The circuit: $|K_7\rangle$ vacuum + V–A vertex on ancillae

The experiment couples the substrate vacuum (the $|K_7\rangle$ graph state on 7 qubits) to two ancilla qubits playing the roles of muon (qubit 7) and electron (qubit 8), and applies a parameterized V–A weak vertex between the ancillae. The substrate-physics narrative is: the muon decays via the V–A chiral interaction (substrate’s $\text{Cl}(0, 7)$ chirality structure exposed by the electroweak shim [16]); the $|K_7\rangle$ vacuum is preserved through the interaction.

The circuit, in pseudocode:

1. Prepare $|K_7\rangle$ on qubits 0–6: apply H to each vertex, then $CZ_{u,v}$ for each of the 21 K_7 edges.
2. Initialise the muon ancilla in $|1\rangle$ (an X gate on qubit 7); the electron ancilla starts in $|0\rangle$.
3. Apply $RXX(\theta) RYY(\theta)$ between qubits 7, 8. This Trotterized V–A mixer implements $\exp[-i\theta(XX + YY)/2]$, which is a partial SWAP on the $\{|10\rangle, |01\rangle\}$ subspace, i.e., a fractional muon-to-electron transfer at angle θ .
4. Measure all 9 qubits.

The total circuit is 9 qubits, 40 gates, depth 13, parameterized in $\theta \in [0, \pi]$ for batched submission. The expected (noiseless) muon-survival probability is $P_{\text{theory}}(\theta) = \cos^2(\theta)$, with $P(0) = 1$, $P(\pi/2) = 0$, $P(\pi) = 1$ (full transfer back at $\theta = \pi$ via the 4π -periodicity of XY rotations).

5.2 Execution on `ibm_marrakesh` (2026-05-01)

The circuit was submitted to `ibm_marrakesh` on 2026-05-01 as job `d7qgmo4f3ras73b5orqg`, batched over 16 angle bindings $\theta \in \{0, \pi/16, 2\pi/16, \dots, \pi\}$ at 4096 shots per binding. Total wall-clock time including queue and execution: 41.5 seconds. Total quantum compute time consumed from the open-instance budget: approximately 11 seconds, well within the 10-minute monthly allocation. The job ran on 9 contiguous physical qubits of `ibm_marrakesh` after `qiskit`’s `transpile` routed the circuit to the device’s heavy-hex coupling map. No mid-circuit measurement; final-state readout only.

5.3 Results: $P(\theta) = 0.963 \cos^2(\theta) + 0.020$

After parsing the per-angle counts to extract the muon-ancilla single-qubit marginal, a least-squares fit to the noise-envelope ansatz $P(\theta) = A \cos^2(\theta) + B$ yields

$$P_{\text{fit}}(\theta) = 0.963 \cos^2(\theta) + 0.020, \tag{10}$$

with $A = 0.963$ (contrast, 96.3% of theory) and $B = 0.020$ (decoherence floor). The plateau values at the endpoints are $P(0) = 0.9946$ and $P(\pi) = 0.9924$, both consistent with $A + B \approx 0.983$ within shot noise. The trough at $\theta = \pi/2$ comes out at $P = 0.027$ instead of theory's 0, the expected decoherence drift toward equilibration through a 40-gate, depth-13 circuit. RMS deviation from theory across the 16 angle points is 1.6% absolute, with maximum deviation 3.3% at intermediate angles. Full data: `analysis/output/heron_runs/exp10_muon_decay_2026-05-01.json`.

5.4 What this verifies about the substrate

Three substrate-monism claims are verified by this run. First, *the $|K_7\rangle$ vacuum is preserved through a 40-gate circuit*: at $\theta = 0$ and $\theta = \pi$, when the V–A vertex is a (near-)identity, $P(\text{muon}) > 0.99$, well above the decoherence threshold. The substrate vacuum is robust at the hardware level. Second, *the V–A weak vertex behaves as predicted by the electroweak shim* on the $|K_7\rangle$ vacuum: the cosine-squared survival curve emerges at 96.3% contrast, matching the substrate-derived prediction without tuning. Third, and most importantly for the thesis of this paper: *the K_7 that hosts the Paper 17 gravitational Wilson amplitude is the same K_7 that runs as a 7-qubit graph state on real superconducting hardware*.

It is important to be careful about what this run does and does not establish. The literal experimental claim is bounded. The $|K_7\rangle$ stabilizer circuit on `ibm_marrakesh` prepares a 7-qubit graph state with the K_7 adjacency, and the $RXX(\theta) \cdot RYY(\theta)$ vertex applied to two ancilla qubits produces a $\cos^2(\theta)$ survival curve at 96.3% contrast through a depth-13 circuit. That is a graph-state benchmark on real superconducting hardware, executed end-to-end from a `nwt_substrate.heron` script. It is *not*, on its own, a verification of the substrate-monism thesis at the ontological level: a hostile referee can fairly say that the $\cos^2(\theta)$ response is the expected output of any parameterized two-qubit entangling vertex applied to a seven-qubit cluster state, and that this fact alone does not identify the IBM circuit with the substrate vacuum of the universe. We accept that critique on its narrow terms.

The reason the run is nevertheless evidentially interesting is *cross-sector*. The same K_7 object—seven vertices, 21 edges, fixed by no tunable parameter—appears in three otherwise unrelated places in the program: the Spin(7) Chern–Simons Wilson amplitude that fixes the gravitational m_e/m_{Pl} (Paper 17), the topology that organizes Paper 6's torus-knot quantum numbers, and the 7-qubit graph state on `ibm_marrakesh`. None of those three uses was tuned to agree with the others. The Heron run does not prove that they are the same physical object; it shows that the same combinatorial object plays a coherent role across mathematically and physically disjoint regimes, and supplies a falsifiable hardware probe (Heron Exp 11 sidereal A/B/C and the cross-laboratory noise correlation program of Section 7.5) of whether that coherence extends to the bulk substrate fields predicted by the program's cosmogenesis hypothesis. In short: the run validates a hardware realization of K_7 at 96.3% contrast; the substrate-monism reading of the run is the *interpretive* hypothesis to which that hardware result speaks.

6 The completed master table

Section 4 ran one electron through seven shims; Section 5 verified one substrate object on real hardware. The full master table assembles every observable the `nwt_substrate` library currently computes from substrate primitives, across all seven sub-disciplines plus the hardware sector, indexed by code reference. This is the substrate-monism demonstration in extenso (Table 4).

Every row carries a code reference of the form `module.fn` or `tests/test_X.py`; running the corresponding library call reproduces the row from substrate primitives (α , m_e , the K_7 adjacency, $\text{Cl}(0,7)$ multiplication, etc.) with no discipline-specific tuning.

Observable	NWT prediction	Status / source	Type
<i>Matter sector (Papers 6, 13, 14)</i>			
24-particle mass spectrum	Paper 6 mass formula, 0 free params	0.76% median residual; nwt.particles	P6
$1/\alpha$	$25\pi\sqrt{3} + 1$ (Paper 8a AB phase)	137.035, 7.6 ppm vs. CO-DATA	I
SM hadron charges	Extended Gell-Mann–Nishijima	38/38 match; nwt.particles.charge	I
<i>QED sector (nwt.qed)</i>			
Compton $ \mathcal{M} ^2$	Substrate Cl(1,3) γ -trace	matches Klein-Nishina to $1:10^{15}$	R
QED β_0 per Dirac fermion	$\frac{2}{3}Q^2N_c$	PDG match; vacuum.polarization	R
$e^+e^- \rightarrow \mu^+\mu^-$	substrate tree-level	matches QED to 10^{-9} at 30 GeV	R
<i>QCD sector (nwt.qcd)</i>			
QCD $\beta_0(n_f = 5)$	23/3	PDG match; nwt.qcd	R
$gg \rightarrow gg$ angular distribution	substrate via BRST identity	matches PDG to machine precision	R
$qq \rightarrow qq, q\bar{q} \rightarrow q\bar{q}$	substrate SU(3) color factors	textbook agreement, 10^{-9}	R
<i>Electroweak sector (nwt.electroweak)</i>			
$\sigma_{\text{peak}}(e^+e^- \rightarrow \mu^+\mu^-)$ at Z	1985 pb (substrate $\gamma + Z$ + interf.)	PDG-LO ≈ 2000 pb (99.2%) [†]	I
Γ_Z (total)	2.422 GeV (LO V-A)	PDG-LO 2.495 GeV (97.1%) [†]	I
<i>String sector (nwt.string)</i>			
K_7 Wilson tower	$n = 16$ (Z), 18 (nucleon), 20 (down), 21 (e^-)	4 SM-scale clean hits	P14
$\alpha^{n/2} m_{\text{Pl}}$	$(p, q) = (2, 1)$	$p^2 + q^2 = 5$ matches Paper 6 prefactor	C
<i>Gravitational sector (nwt.gravity, Paper 18 [17])</i>			
m_e/m_{Pl} (NNLO)	$(8/7)\alpha^{21/2}(1 + \alpha/7 + (21/8)\alpha^2)$	−5.6 ppm vs. CODATA	P17
G (Newton, NNLO)	$(8/7)^2\alpha^{21}$ bracket ² $\hbar c/m_e^2$	−11 ppm vs. CODATA, inside ± 22 ppm error	C
m_{Pl}^2/m_e^2 ($\sim 10^{45}$)	$\alpha^{-21}(8/7)^{-2}$ bracket ^{−2}	+11 ppm vs. CODATA, generated by K_7 alone	C
Schwarzschild as vacuum solution	all 16 $R_{\mu\nu} = 0$ symbolically	exact (sympy proof in nwt.gravity.einstein)	P18
$N_{\text{DOF}} = 12\pi$ structural condition	Sakharov + K_7 amplitude bridge	so(10) GUT gives 39, matches to 3.4%	P18
<i>Hardware sector (nwt.heron, IBM Heron R2)</i>			
Heron Exp 1: $ K_7\rangle$ stabilizer check	all $S_v = +1$	run on ibm.marrakesh , all 7 stabilizers consistent	HW
Heron Exp 2: Z -basis joint distribution	uniform on even-parity bit-strings	run, distribution matches	HW
Heron Exp 4: 3-body $\langle Y_u Y_v Y_w \rangle = 0$	null on non-Fano triples	run, 35/35 triples within shot noise	HW
Heron Exp 10: muon decay on $ K_7\rangle$	$P(\theta) = \cos^2(\theta)$	96.3% contrast, 1.6% RMS dev., 2026-05-01	HW

Table 4: The substrate-monism master table. Every row is computed in **nwt_substrate** from the substrate primitives with no discipline-specific tuning. The **Type** column tags the evidentiary class of each row: **I** = *independent prediction* (a structural commitment of the framework that could fail without other rows failing); **R** = *reproduction of a textbook QED/QCD identity* from substrate algebra (consistency check at the substrate–SM mapping¹⁹ level, not a new prediction); **P6/P14/P17/P18** = *inherited* from cited prior NWT paper, exposed through the library; **C** = *algebraic corollary* of another row (e.g. $(m_{\text{Pl}}/m_e)^2$ is the inverse-square of m_e/m_{Pl}); **HW** = direct *hardware measurement*

6.1 What the table demonstrates

The structural argument is cumulative. Any *one* entry of Table 4 could be regarded as a coincidence: the QCD β_0 matches PDG, but so do many beyond-Standard-Model proposals. G within experimental error of CODATA, but a two-parameter fit can hit any single number. Even m_e/m_{Pl} at 5.6 ppm is dismissable as a numerical fine-tuning. The argument the table is meant to support is the *cumulative* one, but with two important caveats made explicit by the **Type** column.

First, the 22 rows are not 22 independent predictions. Six are textbook QED/QCD identities (**R**) that the substrate $\text{Cl}(0, 7) \rightarrow \text{Cl}(1, 3)$ algebra is computing correctly: they are evidence that the substrate-to-SM *mapping* works at the algebraic level, not new physics predictions. Five rows are inherited from earlier papers (**P6/P14/P17/P18**) and exposed here through the library; they carry the evidentiary weight assigned to them in those papers. Two rows are algebraic corollaries (**C**) of others. The genuinely independent structural predictions plus direct hardware measurements together total nine rows (**I + HW**): $1/\alpha$ to 7.6 ppm, 38/38 Standard-Model hadron charges, the σ_{peak} and Γ_Z at the Z pole, four Heron experiments on `ibm_marrakesh`, and (taken in this paper as a single combined claim with its Paper 17 companion) the m_e/m_{Pl} closed form whose corollaries fix G and the Planck–electron hierarchy.

Second, the cumulative argument the paper makes is *computational*, not statistical: one library, one set of substrate primitives ($\text{Cl}(0, 7)$, $2I$, K_7 , $\text{Spin}(7)$, plus α and m_e as numerical inputs), and no row-specific tuning. The library `nwt_substrate` *exists*; its 344 unit tests *pass*; the module imports show that every shim depends on `nwt.algebra` and `nwt.particles`; the Heron Exp 10 run is on the public IBM Quantum job record. A reader who rejects substrate monism as a metaphysical reading still has to account for the operational fact that one $\sim 5,000$ -line codebase reproduces the 22 rows from a small set of shared primitives. Whether that operational unification reflects an underlying physical substrate or a sophisticated common interface is the central interpretive question of the program; the present paper takes the former hypothesis seriously and provides the executable evidence on which it stands or falls.

The following section 7 catalogs the open problems—points where the table is currently silent, or where the underlying derivation has known gaps. The conclusion (Section 8) returns to the rhetorical implications of the master table for the broader program.

7 Open problems

7.1 The cosmological-constant fine-tuning

The Sakharov-induced gravitational sector (Paper 18) generates a matter-loop contribution to the cosmological constant of order $\Lambda_{\text{cc}} \sim m_{\text{Pl}}^4$, approximately 10^{120} times the observed value $\Lambda_{\text{cc}}^{\text{obs}} \approx 3 \times 10^{-47} \text{ GeV}^4$. This is the standard cosmological-constant problem; it is not specific to NWT, but the substrate-monism framing suggests a partial closure path through Coleman–Weinberg-type cancellations on the same K_7 structure that bridges UV/IR for $1/G$. This is the line of work pursued in Paper 21, deferred from this paper.

7.2 Black hole formation from vortex collapse

Paper 18 verified Schwarzschild as a vacuum solution and established that no fundamental NWT particle is itself a black hole (the Schwarzschild radius of the electron is $\sim 10^{-57} \text{ m}$, far below any substrate scale). An open question is whether dynamical collapse of an NWT vortex configuration to a black hole admits a substrate-resolved description distinct from the Sakharov long-wavelength limit. This is the motivation for Paper 22 (cosmogenesis from a parent black hole), where the substrate-anisotropy axis identified in Section 7.4 is given a dynamical origin.

7.3 The internal $SU(2)$ -vector vs. chiral $SU(2)_L$ puzzle

This is the program’s most serious unresolved structural gap and deserves an explicit statement. The substrate’s internal $SU(2)$ factor (one-third of the 21-dimensional $\mathfrak{so}(7)$ adjoint) acts as a Lorentz *vector* on the $Cl(0,7)$ spinor, whereas the Standard Model’s $SU(2)_L$ acts only on the left-handed chiral component of the Dirac spinor. The library currently handles this at the electroweak-shim level by introducing the V–A vertex factor as a $\gamma^\mu(1 - \gamma_5)/2$ projector applied to the substrate’s $\bar{\psi}\gamma^\mu\psi$ current. We state plainly that *this projector is at present an input to the framework, not a derived consequence of the substrate algebra*. The Heron Exp $10 \cos^2(\theta)$ curve confirms that the V–A projector, once inserted, is empirically correct on hardware, but it does not derive the projector.

The structural shape of the puzzle is mild rather than fatal. The chirality operator γ^5 is constructed from the substrate’s $Cl(1,3)$ generators; the Wick rotation that takes $Cl(0,7) \rightarrow Cl(1,3) \otimes \mathfrak{su}(2)$ selects three of the seven octonionic generators as the internal $\mathfrak{su}(2)$. What is missing is a structural reason for the V–A *contraction* between the chirality projector and the internal $\mathfrak{su}(2)$: why the surviving electroweak gauge boson couples to $\bar{\psi}_L \gamma^\mu T^a \psi_L$ rather than to the parity-symmetric current. Three structural routes are under investigation: (i) chirality emerging from boundary effects of the trefoil background on the Heegaard torus ($\partial T^2 \subset S^3/2I$ vs. bulk dynamics); (ii) parity violation as a Spin(8) triality-breaking phase (left-spinor and right-spinor inequivalent under triality selection); (iii) a K_7 -internal mechanism in which the asymmetry between Eulerian and non-Eulerian circuits selects a chiral sector. None of (i)–(iii) is closed in the present paper; they constitute the principal target of Paper 20.

The fact that the V–A projector is an input rather than a derivation does not affect the master-table numerical predictions (which use it as a fixed ingredient), but it does mean that electroweak chirality is the one place where substrate monism currently bottoms out at an unfilled premise. Closing this gap is the program’s most important next structural deliverable.

7.4 Substrate-axis direction

The cosmogenesis hypothesis examined in Paper 22—that our observable universe nucleated inside a black hole in a parent universe—predicts that the bulk substrate fields of our universe inherit a preferred direction from the host (“parent”) black hole’s accretion-disk angular momentum. Combining this direction with the structural anisotropy required by the $SO(8) \rightarrow Spin(7) \rightarrow 3+1$ holonomy reduction predicts that several independent sky-fixed observables share a single axis: the FLRW null-test direction, the CMB axis-of-evil low- ℓ alignment, and the direction picked out by the recently observed transdimensional anomalous Hall effect (TDAHE) in free-floating thin films [20]. The TDAHE observation is particularly suggestive in the present context: Wang *et al.* note that no existing band-structure or many-body framework predicts a Hall response in two *perpendicular* magnetic fields simultaneously, and that the effect requires three independent broken symmetries operating along orthogonal directions. The $SO(8) \rightarrow Spin(7) \rightarrow SO(7) \rightarrow SO(3,1)$ holonomy chain that organises the substrate’s transverse degrees of freedom supplies precisely three structural symmetry-reduction steps along three substrate-fixed axes; identifying these with Wang *et al.*’s three observed broken symmetries gives a candidate substrate-level explanation for what they note is currently unexplained. Falsifying joint prediction: all sky-fixed anisotropy axes (CMB low- ℓ alignment, Heinesen-Clifton FLRW $Q < 0$ direction, free-floating-thin-film TDAHE direction relative to the lab frame) align in heliocentric coordinates. Heron Experiment 11 (sidereal A/B/C, scaffolded in `nwt_substrate.heron`) is the cheap lab test; the IBM Cloud’s eu-de region availability for pay-go users provides the on-demand continent-spanning baseline. This is one of the most immediate empirical falsification targets for the substrate-monism program.

7.5 Falsification path

Substrate monism, as a computational hypothesis about a single shared library that reproduces a master table, would be most straightforwardly falsified by a master-table row that the library cannot reproduce. But the more interesting falsifiers are the ones that are not in the master table: predictions the program makes that no other framework predicts in the same sharply-correlated form, where a negative observation is fatal to the substrate-monism reading without being fatal to any particular individual derivation. Three such predictions are already concrete enough to test, and a fourth is library-driven.

F1. Substrate-axis joint correlation. As argued in Section 7.4, the cosmogenesis + holonomy-reduction story predicts that three independent sky-fixed anisotropy axes should align in heliocentric coordinates: the FLRW null-test direction (Heinesen–Clifton), the CMB axis-of-evil low- ℓ alignment, and the lab-frame Hall direction of the transdimensional anomalous Hall effect (TDAHE) in 2–5 nm rhombus-pattern free-floating thin films [20]. Each has an independent observational claim of an anisotropy axis at the $\sim 2\text{--}4\sigma$ level (the cosmological pair) or as a discovered laboratory effect (TDAHE); substrate monism predicts they are the same axis, with directional cosines agreeing to within the respective measurement uncertainties. The TDAHE leg is the cheapest of the three to test: where the cosmological axes require survey-data reanalysis, the TDAHE direction can be measured by repeating the Wang *et al.* setup with the film orientation tracked in heliocentric coordinates, an experiment of order weeks rather than years. An observation that the three axes are uncorrelated, or correlated with directions unrelated to a stable astrophysical referent, falsifies the substrate-axis hypothesis. No other current framework predicts this joint correlation; in particular, Wang *et al.* note that no existing theory predicts the TDAHE itself, so the substrate-monism reading both supplies the missing theoretical structure and stakes a falsifiable directional commitment on it.

F2. Heron Experiment 11 sidereal A/B/C. The same substrate-axis prediction, on the lab side, predicts that the K_7 stabilizer-state coherence times measured on `ibm_marrakesh` should depend on the device’s heliocentric orientation at run time, with a sidereal modulation period (23 h 56 m 4 s) and phase determined by F1. The experiment is scaffolded in `nwt_substrate.heron (exp11_sidereal)` as three batched submissions A/B/C separated by 8-hour sidereal phase offsets, each preparing $|K_7\rangle$ and measuring the stabilizer-product expectation as a coherence proxy at 4096 shots per binding. Total quantum-compute budget: ~ 30 seconds per phase point, ~ 3 minutes for a full A/B/C scan, well within the open-instance monthly allocation. A null result—A/B/C coherence values consistent within shot noise—falsifies the sidereal-coupling reading of the substrate-axis claim independently of F1’s astrophysical channels.

F3. K_7 Wilson-tower predictions at unanchored n . Paper 14’s K_7 Wilson tower predicts $\alpha^{n/2} m_{\text{Pl}}$ at $n = 1, \dots, 21$. Four anchors land at SM-scale observations: $n = 16$ at M_Z , $n = 18$ at the nucleon, $n = 20$ at the down quark, $n = 21$ at the electron. The remaining seventeen n -values predict mass scales which, if they correspond to physical resonances, should be observable as substrate-only states (no SM gauge couplings, but graviton-like coupling to bulk geometry). The tightest pair of predictions sits at $n = 8, 13$ (TeV-PeV scale), which is within reach of present collider data with a substrate-specific search strategy.

F4. Hopf-linked exotic-hadron predictions. The connected-sum / Hopf-link composition law on knot topology generates a closed catalog of ~ 395 predicted exotic-hadron states across 12 flavor sectors, each with a structural mass-scale prediction from Paper 6’s torus-knot mass formula applied to the connected sum. These are stored in the `compendium_extended` module of `nwt.particles`

and are individually testable by future hadronic searches. Discovery rates significantly below ~ 1 in the predicted catalog would disfavor the K_7 -organized hadron-spectrum reading without disfavoring substrate monism in general; mass-scale offsets larger than the Paper 6 residual ($\sim 0.76\%$ median) on several entries would be a sharper signal.

Schedule. F2 is the program’s nearest-term falsifier. The three sidereal phase points require only that the IBM Quantum job submission be timed to the host facility’s local sidereal time; the phase binning is set by Qiskit’s job-submission timestamp, not by any device-side calibration. Results, positive or null, will be reported in a future paper covering the next batch of Heron experiments. F1’s astrophysical correlation analysis depends on third-party survey data (CMB low- ℓ alignment from Planck PR4, the Heinesen–Clifton FLRW null-test catalog, and follow-up TDAHE measurements with helio-centric orientation tracking); the joint-axis correlation will be reported in a future paper on cosmogenesis, with the substrate-axis fit performed in `nwt_substrate.gravity.substrate_axis`. F3 (K_7 Wilson tower at unanchored n) requires a substrate-specific reanalysis of LHC Run-3 high-mass dijet and dimuon spectra in the $n = 8$ ($\sim \text{TeV}$) and $n = 13$ ($\sim \text{PeV}$ -impact-parameter-equivalent) regions, scoped for a future paper. F4 (Hopf-linked exotic catalog) is a standing prediction against future hadronic searches and has no internal program timeline; the catalog is published in the `compendium.extended` module of `nwt.particles` for any group that wishes to test it.

These four predictions are not equally weighted. F1 and F2 are the program’s primary falsifiers because they predict a *joint* pattern that no other current framework predicts; F3 and F4 are extensions of derivations already in the master table to regimes where the framework has not yet been empirically anchored. None of the four requires accepting substrate monism at the ontological level to be tested.

8 Conclusion

The substrate-monism thesis is, at base, a re-identification: the program proposes that the same set of mathematical objects—octonion Clifford $\text{Cl}(0, 7)$, binary icosahedral $2I$, the Brieskorn–Poincaré sphere $S^3/2I$, and the complete graph K_7 on its Heegaard torus—can serve as a common ontology for the seven sub-disciplines of the Standard Model and gravity, with each sub-discipline obtained as a particular reading of those objects under its own community-standard vocabulary. The thesis is operationalized through the `nwt_substrate` library, which commits to specific Python objects, applies the derivations of Papers 6 and 13–18 to those objects, and reports back what the seven view-shims compute.

We have shown that this implementation reproduces the master table. Table 4 catalogs twenty-two rows across seven sub-disciplines—matter spectrum, α , hadron charges, QED and QCD amplitudes and β -functions, electroweak Z-pole physics, the K_7 Wilson tower, the gravitational G and m_e/m_{Pl} closed forms and Schwarzschild, hardware K_7 stabilizers, and the Heron Exp 10 muon decay—all computed from the same substrate primitives in $\sim 5,000$ lines of Python with 344/344 unit tests passing. The Type-classified breakdown (Section 6, Table 4) distinguishes 5 independent structural predictions and 4 direct hardware measurements from the 13 inherited or mapping-consistency rows; the substrate-monism reading rests on the joint coherence of these classes plus the operational fact that one library, with no row-specific tuning, reproduces all twenty-two.

The companion Paper 18 [17] supplies the gravitational column at the dynamical level (Sakharov-induced Einstein equations from the BPS matter sector with the K_7 -fixed cutoff; Schwarzschild verified symbolically; G to -11 ppm of CODATA, inside the ± 22 ppm experimental band). Together with Papers 6 and 13–17 supplying the Standard-Model columns, the table is, as a first integration pass, jointly populated by the program’s existing derivations.

The remaining work is twofold. First, the open structural loops—the chiral- $\text{SU}(2)_L$ derivation, the

cosmological-constant fine-tuning, and the structural origin of the substrate-axis direction—are the program’s principal internal asks; they are queued as Papers 20–22. Second, and more important, the framework now exposes a small number of empirical falsifiers (Section 7.5, predictions F1–F4) that no other framework predicts in the same correlated form. The joint sky-fixed alignment of the FLRW null-test, the CMB axis-of-evil, the TDAHE thin-film direction, and the Heron Exp 11 sidereal modulation is the substrate-axis test; the K_7 Wilson tower at unanchored n values is the mass-spectrum test; the Hopf-linked exotic catalog is the hadron-spectrum test. Each is falsifiable at percent-level precision with current technology.

The library is the operational embodiment of the hypothesis: what the paper claims in prose, the library executes in code, and the master table is the public record of that execution. Whether the substrate primitives are ontologically primary or only a particularly compact common interface is the program’s interpretive question; it is not settled by the present paper. What is settled is that one library, four substrate primitives, and two numerical inputs (α , m_e) suffice to reproduce twenty-two rows from seven sub-disciplines without row-specific tuning, and that several falsifying predictions of the substrate-monism reading are now within reach of laboratory and cosmological measurement. Whether the framework survives those tests is the next question.

Acknowledgements

We thank ChatGPT (OpenAI), Gemini (Google), Perplexity, and Claude Mobile for critical review of this paper and earlier papers in the series.

Code availability

The `nwt_substrate` library is released as a separate public Python package in the `nwt-substrate` repository at <https://github.com/JimGalasyn/nwt-substrate>, archived on Zenodo with concept DOI [10.5281/zenodo.20012027](https://doi.org/10.5281/zenodo.20012027) (always-latest) and version DOI [10.5281/zenodo.20041585](https://doi.org/10.5281/zenodo.20041585) for v0.1.3, the version against which this paper’s results were verified. The papers, analysis scripts, and Heron experiment data live in the public NWT monorepo at <https://github.com/JimGalasyn/null-worldtube>. The relevant artifacts are:

- Library `nwt_substrate` (separate repo `nwt-substrate`) v0.1.3 (344/344 tests passing, `python3 -m pytest nwt_substrate/tests/`).
- Multi-view demo (public monorepo): `analysis/nwt_multiview_demo.py`.
- Heron Experiment 10 source and run data (public monorepo):
`analysis/nwt_muon_decay_heron_demo.py`,
`analysis/output/heron_runs/exp10_muon_decay_2026-05-01.json`.
- Gravity shim (in `nwt-substrate`): `nwt_substrate/gravity/`, packaging the Paper 18 [17] derivation; includes the symbolic Schwarzschild proof `verify_schwarzschild_vacuum_symbolic`.
- Paper 18 G1–G6 reproduction scripts (public monorepo): `analysis/nwt_paper18-G{1..6}*.py`.

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